Juan Becerra

Software Engineer – Systems, Tools & Animation

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Experience

Electronic Arts – Lifestyle **2025 – Present**
Software Engineer – Tools & Tech-Art *Kirkland, WA*

- Engineered a high-throughput Unreal Engine C++ plugin exporting runtime character trajectory, pose, and velocity data to train and populate motion-matching query databases.
- Prototyped and integrated ML-assisted authoring pipelines into DCC tools (leveraging PyTorch and ONNX runtime), enabling automated asset tagging and procedural pose generation without disrupting artist control.
- Architected an extensible Maya/Python animation stylization tool adopted across multiple internal studios, reducing redundant manual keyframing workflows by over 70%.

Electronic Arts – Maxis Studios **2021 – 2025**
Software Engineer – Gameplay & Animation *Seattle, WA (Remote)*

- Architected a multi-threaded asynchronous path-planning and navigation API in UE5/C++ for *Project Rene* (*The Sims*), enabling deterministic crowd routing and real-time obstacle avoidance within strict frame budgets (<1.5ms).
- Profiled and optimized runtime memory and frame budgets across character evaluation graphs and query sweeps in a large UE5/C++ codebase, eliminating hitches during dense crowd simulation scenarios.
- Partnered directly with technical animators and gameplay designers to author in-editor tuning tools and visual debuggers, accelerating iteration cycles and authoring loops for complex behavior trees.
- Implemented core locomotion and runtime animation systems in UE5 (motion-matching, distance matching, custom IK solvers), delivering fluid, responsive character controls for next-gen simulation gameplay.

Lionbridge **2020 – 2021**
Senior Games Test Engineer *Boise, ID*

- Led a team of QA testers performing combinatorial and regression testing for live-service AAA titles, authoring technical repro steps and engine-level diagnostics in JIRA to accelerate developer triage.

Projects

Agentic Resume Pipeline | *Claude Code, Gemini, PowerShell, LaTeX* | GitHub **Oct. 2026**

- Built an agent-driven workflow with authored agent skills and scoped context: agents read job postings, tailor LaTeX resumes against an evidence-backed skills inventory, compile, and verify output.

Hoowan Game Engine | *C++, OpenGL, EnTT, CMake* | GitHub **Nov. 2020**

- Engineered a custom C++ game engine with an OpenGL rendering submission pipeline, EnTT-based ECS, input event polling, and 2D collision, plus a companion desktop editor for scene composition.

DoomSlop | *JavaScript, Three.js, Web Audio API, Vite* | Live Demo | GitHub **Sep. 2026**

- Built a real-time 3D browser first-person shooter in JavaScript featuring hitscan raycasting, synthesized Web Audio sound effects, custom canvas radar minimap, and automated CI/CD deployments via GitHub Pages.

Technical Skills

Languages: C++ (Modern, Multi-threading), Python, JavaScript, TypeScript, C#, Java, GLSL/HLSL, Rust, SQL
AI & Agentic Development: Claude Code, Cursor, Codex, Agent Harnesses, Multi-Agent Systems, Agent Evaluation, Context Engineering, PyTorch/ONNX
Engines & Graphics: Unreal Engine 5 (Source, Motion Matching, GAS, Mass/Crowd), Unity, Godot, OpenGL, Vulkan
Tools & Desktop UI: PySide/Qt, ImGui, In-Editor Tooling, Unreal Plugin Development, Maya API, DCC Pipeline Automation
Build & Infrastructure: CMake, GitHub Actions, CI/CD, Git, Docker, Linux, AWS, Visual Studio, Perforce (P4), JIRA
Engineering Practices: Code Reviews, Design Reviews, Unit Testing, Technical Roadmap Planning, Profiling & Optimization

Education & Continuing Studies

Boise State University **2017 – 2021**
Bachelor of Science in Computer Science *Boise, ID*

University of Washington (Continuing Education) **2026 – Present**
Graduate Coursework in Machine Learning & Deep Learning (Non-Degree, In Progress) *Seattle, WA*